

Qianxun (Cecilia) Xu

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EDUCATION

University of California, Los Angeles

2026–2028

M.S. in Computer Science

Duke Kunshan University & Duke University Dual Degree

2022–2026

- B.S. in Applied Mathematics and Computer Science; Computer Science Track
- GPA: 3.93/4.0 | Major GPA: 3.97/4.0
- Awards: IEEE ICPADS Best Presentation in Session; NeurIPS Scholar Award (\$2,000); DKU Merit-based College Entrance Scholarship (\$48,000)

PUBLICATIONS

Xu, Q., Song, C., Cai, Y., Zhang, C. SwitchCraft: Training-Free Multi-Event Video Generation with Attention Controls. *CVPR*, 2026 (main track poster).

Xu, Q., Boukerche, A., Sun, P. FENSE: Feedback-Enabled Neighbor Selection for Spatial Aware Collaborative Perception. *IEEE ICPADS*, 2025 (oral).

Xu, Q., Li, Z. Partial Physics Informed Diffusion Model for Ocean Chlorophyll Concentration Reconstruction. *NeurIPS*, 2025 (main track poster).

INDUSTRY EXPERIENCE

StepFun | Research Intern, World Models

Apr–Sep 2026

ConsistWorld: Evidence Routing for Consistent Multi-Agent World Models

- Designed memory retrieval and visibility-gated sharing with bounded active context, reducing cross-agent appearance inconsistency by **69%** on asynchronous handoffs versus a baseline without either module.
- Built an end-to-end **synthetic-data pipeline** integrating calibrated camera geometry and motion-constrained trajectory generation to produce aligned multi-view videos with diverse revisit patterns for training consistent world models.
- Developed a **staged training curriculum**: stable teacher-forced initialization on general multi-view data, self-resampling to mitigate rollout errors, and targeted adaptation on synthetic revisit data.
- Designed **consistency evaluation protocols** for revisits and cross-agent sharing, validating generalization to unseen agent counts and longer rollouts. Demonstrated improved visual fidelity, reducing perceptual error by **19%** compared with full-history attention.

RESEARCH EXPERIENCE

Westlake University | Undergraduate Researcher

May–Nov 2025

SwitchCraft: Training-Free Multi-Event Video Generation | Advisor: Chi Zhang

- **Problem:** Addressed semantic confusion, temporal misalignment, and high fine-tuning costs in existing text-to-video diffusion models when generating multi-event videos.
- **Method:** Developed **Event-Aligned Query Steering** to strengthen target-event attention and suppress competing events within designated temporal spans. Designed an **Auto-Balance Strength Solver** using convex optimization to balance temporal control and visual fidelity.
- **Results:** Achieved SOTA text-to-video alignment among evaluated methods, improving alignment scores by **24%** over the Wan2.1 T2V-14B baseline while preserving temporal smoothness and visual quality.

Duke Kunshan University | Undergraduate Thesis

Jan–Aug 2025

FENSE: Communication-Efficient Collaborative Perception | Advisor: Peng Sun

- **Problem:** Addressed redundant feature transmission and high communication latency caused by highly overlapping fields of view in V2X collaborative perception.
- **Method:** Grouped overlapping vehicles via traffic-aware dynamic clustering. Leveraged low-bandwidth feedback signals to evaluate cluster representatives, selectively requesting high-dimensional features to eliminate redundancy.
- **Results:** Achieved superior accuracy-efficiency trade-off. Improved detection accuracy by **~4%** under equal bandwidth while significantly reducing ego-side computation time.

Duke Kunshan University | Undergraduate Researcher

May 2024–May 2025

PPIDM: Partial Physics-Informed Diffusion for Ocean Reconstruction | Advisor: Zuchuan Li

- **Problem:** Addressed sparse data imputation in environmental monitoring, overcoming poor generalization in data-driven models and rigid mechanism modeling in traditional physics-based methods.
- **Method:** Designed selective physics encoding based on observable advection dynamics and a subspace alignment loss to align predictions with ground truth within the known physics subspace, avoiding full-equation residual constraints when parts of the dynamics are unknown.
- **Results:** Achieved robust long-range spatiotemporal inference. Reduced reconstruction RMSE by **34%** against pure data-driven models, and by **49%** against baselines forcing complete physical equations.

TEACHING EXPERIENCE

Teaching Assistant, Introduction to Computer Science

Oct–Dec 2022

- Led weekly labs and office hours, graded quizzes and assignments, and analyzed recurring errors to recommend curriculum adjustments, increasing student engagement and academic achievement.

SKILLS & LANGUAGES

Technical: Python, Java, C, PyTorch, Git, Linux, distributed training

Languages: Mandarin (native), English (TOEFL 116/120), German (intermediate)